A corda vibrante



* Corda discreta: $m\ddot{q}_j = \underbrace{m\omega_0^2}_{\uparrow\ K=\frac{T}{d}}\left((q_{j+1}-q_j)-(q_j-q_{j-1})\right)$

* Paso ao continuo

$m\to 0$, $d\to 0$ $\quad \rho=\frac{m}{d}\to$ constante

$jd \longrightarrow x$

$(j-1)d \longrightarrow x-d$

$(j+1)d \longrightarrow x+d$

$q_j(t)\equiv q(jd,t) \longrightarrow q(x,t)$

- Divido por d: $\frac{m}{d}\ddot{q}_j(x) = m\omega_0^2\left(\frac{q(x+d)-q(x)}{d} - \frac{q(x)-q(x-d)}{d}\right)$

- Paso ao continuo tomando o limite $d\to 0$.

$$\rho\ddot{q} = m\omega_0^2\left(\lim_{d\to 0}\frac{q(x+d)-q(x)}{d} - \lim_{d\to 0}\frac{q(x)-q(x-d)}{d}\right)$$

$\frac{\partial q}{\partial x}\Big|_{x=jd}$ $\qquad$ $\frac{\partial q}{\partial x}\Big|_{x=(j-1)d}$

- Sabendo que $m\omega_0^2 = K = \frac{T}{d}$, voltemos a tomar o limite.

$$\rho\ddot{q} = T\lim_{d\to 0}\frac{\left.\frac{\partial q}{\partial x}\right|_x - \left.\frac{\partial q}{\partial x}\right|_{x-d}}{d}$$

$$\Rightarrow \rho\ddot{q} = T\frac{\partial^2 q}{\partial x^2} \longrightarrow \boxed{\frac{\partial^2 q}{\partial x^2} = \frac{1}{v^2}\frac{\partial^2 q}{\partial t^2}}$$ Ec. de Ondas

$v \equiv \sqrt{\frac{T}{\rho}}$

$\uparrow$ $\frac{\partial^2 q}{\partial t^2}$ xa que en $q(x,t)$ x non depende de t.

Lagrauxiana da corda

$$L = \frac{\rho}{2} \sum_{j=1}^{n+1} \frac{1}{2}\left(m \dot{q}_j^2 - m\omega_0^2 (q_i - q_{j-1})^2 \right)$$

multiplico e divido por d.

$$L = \frac{\rho}{2} d \sum_{j=1}^{n+1} \frac{1}{2}\underbrace{\left(\rho \dot{q}_j^2 - T\left(\frac{q(jd,t) - q((j-1)d,t)}{d}\right)^2 \right)}_{L_j}$$

$$L = d \sum_{j=1}^{n+1} L_j$$

Paso ao contínuo

$$\lim_{d\to 0} \frac{q(x,t) - q(x-d,t)}{d} = \frac{\partial q}{\partial x}$$

$$\lim_{d\to 0}\left(d \sum_{j=1}^{n+1} \right) = \int_0^{L=(n+1)d} dx$$

$$\Rightarrow L = \int_0^L dx \underbrace{\frac{1}{2}\left(\rho\left(\frac{\partial q}{\partial t}\right)^2 - T\left(\frac{\partial q}{\partial x}\right)^2 \right)}_{\mathcal{L}}$$

$\mathcal{L}$ → densidade lagrauxiana!

Eq do movimento

Cambio notación $q(x,t) \to \psi(x,t)$ (CAMPO); $\dot{\psi} = \frac{\partial \psi}{\partial t}$; $\psi' = \frac{\partial \psi}{\partial x}$

$$\mathcal{L} = \frac{1}{2}\left(\rho \dot{\psi}^2 - T \psi'^2 \right)$$

Ec. do movimento → $\rho \ddot{\psi} = T \psi''$

Ecuación de Euler-Lagrange para $\mathcal{L}$

$$\mathcal{L} = \mathcal{L}(\psi, \dot{\psi}, \psi', x, t) \; ; \; \psi = \psi(x,t)$$

$$\frac{\partial}{\partial t}\left(\frac{\partial \mathcal{L}}{\partial \dot{\psi}}\right) + \frac{\partial}{\partial x}\left(\frac{\partial \mathcal{L}}{\partial \psi'}\right) = \frac{\partial \mathcal{L}}{\partial \psi}$$

Ver Demo en Marion

análogo a $\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i}\right) = \frac{\partial L}{\partial q_i}$

Lagranxiana relativista

$x \longrightarrow (x_0, \vec{x}) \longrightarrow x_\mu \qquad \mu = 0, 1, 2, 3$

$\uparrow$
ct

$\psi \longrightarrow \psi_\alpha \qquad \alpha = 1, \ldots,$ nº de campos.

$$\frac{\partial \psi_\alpha}{\partial x_\mu} \equiv \psi_{\alpha,\mu}$$

$$\Rightarrow \qquad \mathcal{L} = \mathcal{L}(\psi_\alpha, \psi_{\alpha,\mu}, x_\mu)$$

Ecuacións do movemento:

$$\frac{d}{dx_\mu}\left(\frac{\partial \mathcal{L}}{\partial \psi_{\alpha,\mu}}\right) = \frac{\partial \mathcal{L}}{\partial \psi_\alpha}$$

As coordenadas "xeneralizadas" son "campos" que non teñen porque ter natureza "mecánica".

Por exemplo → Campo EM
función de onda da Mec. cuántica
Campo de velocidades en fluídos
Campos relativistas de partículas